Anesthesia for Endocrinal Disorders MCQ – Chapter 23

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 7: Anesthesia for Coexisting Diseases (Chapter 23)

50 Questions • 30 Minutes

Instructions:
• Each question has 4 options (A, B, C, D).
• The correct answer is shown on the right side of each question.
• Detailed explanations with textbook page references are at the end.

Questions

Q1. What are the basic goals of preoperative evaluation in a diabetic patient? **A.** Only checking blood sugar levels **B.** Assess severity by blood sugar & HbA1c, rule out autonomic and peripheral neuropathy, and organ damage **C.** Only ruling out nephropathy **D.** Only checking HbA1c levels	**Ans: B**
Q2. Why should premedication against aspiration be given to diabetic patients? **A.** They have increased gastric acid production **B.** They have delayed gastric emptying **C.** They are on multiple medications **D.** They have esophageal dysfunction	**Ans: B**
Q3. How long before surgery should sulfonyl urea and metformin be stopped? **A.** On the morning of surgery **B.** 12 hours before **C.** 24–48 hours prior to surgery **D.** 1 week before surgery	**Ans: C**
Q4. In the traditional insulin regime, what fraction of the AM dose of insulin is given on the morning of surgery? **A.** 1/4 of the AM dose **B.** 1/3 of the AM dose **C.** 1/2 of the AM dose **D.** Full AM dose	**Ans: C**
Q5. At what rate should 5% dextrose be infused when giving insulin on the morning of surgery? **A.** 50 mL/hour **B.** 100 mL/hour **C.** 125 mL/hour **D.** 150 mL/hour	**Ans: C**
Q6. By how much does 1 unit of regular insulin decrease blood sugar? **A.** 10–15 mg/dL **B.** 15–20 mg/dL **C.** 25–30 mg/dL **D.** 40–50 mg/dL	**Ans: C**

Q7. What is the formula for regular insulin infusion rate (units/hour) as per current guidelines? **A.** Plasma glucose (mg/dL) / 100 **B.** Plasma glucose (mg/dL) / 150 **C.** Plasma glucose (mg/dL) / 200 **D.** Plasma glucose (mg/dL) / 250	**Ans: B**

Q8. What is the target plasma sugar level during the perioperative period? **A.** 100–130 mg/dL **B.** 130–150 mg/dL **C.** 150–180 mg/dL **D.** 180–220 mg/dL	**Ans: C**

Q9. What is the most common cause of death in the perioperative period in diabetic patients? **A.** Hypoglycemia **B.** Diabetic ketoacidosis **C.** Myocardial ischemia **D.** Renal failure	**Ans: C**

Q10. Why may MI go silent in the postoperative period in diabetic patients? **A.** Due to analgesics masking pain **B.** Due to autonomic neuropathy **C.** Due to sedation **D.** Due to peripheral neuropathy	**Ans: B**

Q11. Which patients with diabetes can safely receive regional anesthesia? **A.** All diabetic patients **B.** Patients not having peripheral and/or autonomic neuropathy **C.** Only type 2 diabetic patients **D.** Only insulin-dependent patients	**Ans: B**

Q12. Why may difficult intubation be anticipated in diabetic patients? **A.** Due to obesity **B.** Due to involvement of atlanto-occipital and temporomandibular joint (stiff joint syndrome) **C.** Due to macroglossia **D.** Due to tracheal compression	**Ans: B**

Q13. What may act as a savior when hypotension and bradycardia in diabetic patients with autonomic neuropathy do not respond to atropine and ephedrine? **A.** Dopamine **B.** Noradrenaline **C.** Intravenous adrenaline **D.** Phenylephrine	**Ans: C**

Q14. Which of the following drugs should be avoided in diabetic patients because they cause sympathetic stimulation and hyperglycemia? **A.** Propofol, Vecuronium, Isoflurane **B.** Ketamine, Pancuronium, Atropine, Desflurane >6% **C.** Thiopentone, Sevoflurane, Glycopyrrolate **D.** Etomidate, Atracurium, Fentanyl	**Ans: B**

Q15. When should elective surgery be performed in a hyperthyroid patient? **A.** Immediately with beta-blocker cover **B.** After the patient is euthyroid **C.** After 2 weeks of antithyroid drugs **D.** Only after total thyroidectomy	**Ans: B**

Q16. What should be brought down to normal with beta blockers before taking a hyperthyroid patient for surgery? **A.** Blood pressure **B.** Sleeping pulse rate **C.** Thyroid hormone levels **D.** Temperature	**Ans: B**

Q17. Why should airway assessment be done in hyperthyroid patients with large thyroid? **A.** Risk of vocal cord paralysis **B.** Large thyroid can produce tracheal compression **C.** Risk of aspiration **D.** Difficulty in mask ventilation	**Ans: B**

Q18. Why should local anesthetics with adrenaline NOT be used in hyperthyroid patients? **A.** They are less effective **B.** They can cause sympathetic stimulation **C.** They are metabolized faster **D.** They cause hypothermia	**Ans: B**

Q19. Why are central neuraxial blocks preferred in hyperthyroid patients? **A.** They provide better analgesia **B.** They block the sympathetic response **C.** They are easier to administer **D.** They reduce thyroid hormone levels	**Ans: B**

Q20. Which is the induction agent of choice in hyperthyroid patients and why? **A.** Propofol – fast onset **B.** Thiopentone – possesses antithyroid property **C.** Etomidate – cardiovascular stability **D.** Ketamine – bronchodilator	**Ans: B**

Q21. What is the preferred maintenance agent for hyperthyroid patients prone to arrhythmias? **A.** Halothane **B.** Sevoflurane **C.** Isoflurane **D.** Desflurane	**Ans: C**

Q22. Does the MAC of inhalational agents change in hyperthyroidism? **A.** Yes, it increases significantly **B.** Yes, it decreases significantly **C.** No, MAC remains the same **D.** It varies depending on the agent	**Ans: C**

Q23. What is a thyroid storm and when does it most often occur? **A.** A benign condition occurring preoperatively **B.** A life-threatening condition most often occurring in postoperative period **C.** An allergic reaction during induction **D.** A transient condition during maintenance	**Ans: B**

Q24. Which drug is used in thyroid storm to block conversion of T4 to T3? **A.** Hydrocortisone **B.** Dexamethasone **C.** Prednisolone **D.** Methylprednisolone	**Ans: B**

Q25. What is the most common cause of upper airway obstruction after thyroid surgery? **A.** Laryngeal edema **B.** Tracheal compression by expanding hematoma **C.** Vocal cord paralysis **D.** Tracheomalacia	**Ans: B**

Q26. What can hypoparathyroidism after thyroidectomy cause? **A.** Hyperkalemia **B.** Hypocalcemia leading to laryngospasm **C.** Hypernatremia **D.** Metabolic alkalosis	**Ans: B**

Q27. What is the emergency treatment of choice for a hyperthyroid patient undergoing emergency surgery? **A.** IV hydrocortisone only **B.** IV esmolol or propranolol to settle tachycardia **C.** Oral antithyroid drugs only **D.** Plasmapheresis	**Ans: B**

Q28. Should elective surgery be deferred in all hypothyroid patients? **A.** Yes, in all cases **B.** No, mild hypothyroidism should not be a deterrent to postpone the surgery **C.** Only in patients above 60 years **D.** Only in patients with goiter	**Ans: B**
Q29. Which of the following are associated conditions with hypothyroidism that should be ruled out preoperatively? **A.** Hypertension, tachycardia, hyperthermia **B.** Congestive cardiac failure, anemia, hypothermia, hyponatremia, hypoglycemia **C.** Hyperglycemia, hyperkalemia, polycythemia **D.** Pulmonary hypertension, renal failure	**Ans: B**
Q30. Why should premedication with benzodiazepines be avoided in hypothyroid patients? **A.** They are metabolized faster **B.** They cause hypothermia **C.** These patients are extremely sensitive to depressant drugs **D.** They cause hyponatremia	**Ans: C**
Q31. What is the induction agent of choice in hypothyroid patients with normal cardiac output? **A.** Thiopentone **B.** Propofol **C.** Etomidate **D.** Ketamine	**Ans: B**
Q32. Which muscle relaxants are preferred in hypothyroid patients? **A.** Pancuronium and Vecuronium **B.** Mivacurium, Atracurium/Cis-atracurium **C.** Succinylcholine **D.** Rocuronium	**Ans: B**
Q33. Why should all drugs be used in minimal possible doses in hypothyroid patients? **A.** They have increased renal clearance **B.** They have increased hepatic metabolism **C.** The metabolism of drugs is reduced in hypothyroidism **D.** They have increased protein binding	**Ans: C**
Q34. What additional cover is needed for emergency surgery in severely hypothyroid patients? **A.** IV antibiotics only **B.** IV T3 and T4, and steroid cover **C.** IV calcium only **D.** IV magnesium only	**Ans: B**

Q35. What drug is used to stabilize hypertension preoperatively in pheochromocytoma? **A.** Propranolol **B.** Atenolol **C.** Phenoxybenzamine (alpha blocker) **D.** Nifedipine	**Ans: C**

Q36. Why must beta blockers NOT be started before alpha blockers in pheochromocytoma? **A.** Beta blockers are ineffective alone **B.** Unopposed alpha action can cause malignant hypertension **C.** Beta blockers cause hyperglycemia **D.** Alpha blockers take longer to act	**Ans: B**

Q37. Why is halothane absolutely contraindicated in pheochromocytoma? **A.** It causes bronchospasm **B.** It sensitizes myocardium to catecholamines **C.** It causes hepatotoxicity **D.** It causes malignant hyperthermia	**Ans: B**

Q38. What can happen during tumor handling in pheochromocytoma surgery? **A.** Hypothermia **B.** Profound hypertension and life-threatening arrhythmias due to increased catecholamines **C.** Hypoglycemia only **D.** Bradycardia only	**Ans: B**

Q39. What is the most common cause of death in the immediate postoperative period in pheochromocytoma patients? **A.** Arrhythmia **B.** Hypertensive crisis **C.** Hypotension **D.** Hypoglycemia	**Ans: C**

Q40. What should be started immediately after tumor vein ligation in pheochromocytoma surgery? **A.** Normal saline **B.** Ringer lactate **C.** Dextrose normal saline **D.** Colloids	**Ans: C**

Q41. For how long should hydrocortisone be continued after unilateral adrenalectomy for pheochromocytoma? **A.** 12 hours **B.** 1–2 days **C.** 3–6 days **D.** 1 week	**Ans: B**

Q42. What effects of cortisol must be corrected before elective surgery in Cushing syndrome? **A.** Only hypertension **B.** Hypertension, hyperglycemia, volume overload, and hypokalemia **C.** Only hyperglycemia **D.** Only hypokalemia	**Ans: B**

Q43. What is the most common cause of secondary adrenal insufficiency? **A.** Pituitary tumor **B.** Autoimmune adrenalitis **C.** Exogenous steroids **D.** Tuberculosis	**Ans: C**

Q44. Any patient who has taken steroids (prednisone >5 mg/day) for more than how many weeks in the last 1 year must receive intraoperative steroid supplementation? **A.** 1 week **B.** 2 weeks **C.** 3 weeks **D.** 4 weeks	**Ans: C**

Q45. What is the dose of hydrocortisone supplementation for patients with adrenal insufficiency? **A.** 50 mg 8 hourly **B.** 100 mg 8 hourly **C.** 200 mg 12 hourly **D.** 50 mg 12 hourly	**Ans: B**

Q46. Why is etomidate contraindicated in adrenal insufficiency? **A.** It causes hypotension **B.** It causes adrenocortical suppression **C.** It causes tachycardia **D.** It is poorly metabolized	**Ans: B**

Q47. Why should sevoflurane be avoided in mineralocorticoid excess with hypokalemic nephropathy? **A.** It causes bronchospasm **B.** It produces fluoride that can worsen nephropathy **C.** It causes hyperkalemia **D.** It causes tachycardia	**Ans: B**

Q48. What is the most important concern for acromegaly patients undergoing anesthesia? **A.** Hyperglycemia **B.** Difficult airway management **C.** Cardiac arrhythmias **D.** Renal dysfunction	**Ans: B**

Q49. How long does recovery of hypothalamic-pituitary axis (HPA) take after exogenous steroid use? **A.** 3 months **B.** 6 months **C.** 1 year **D.** 2 years	**Ans: C**

Q50. What can happen in hypothyroid patients in the postoperative period due to hypoventilation? **A.** Hyperglycemia **B.** Hypoxia **C.** Hypertension **D.** Tachycardia	**Ans: B**

Answer Key & Explanations

Q1. Answer: B — Assess severity by blood sugar & HbA1c, rule out autonomic and peripheral neuropathy, and organ damage

The basic goals are to assess severity by checking blood sugar and HbA1c, rule out autonomic neuropathy, peripheral neuropathy, and organ damage especially nephropathy.

Topic: Diabetes Mellitus – Preoperative Evaluation | Ref: Ch 23, Page 206

Q2. Answer: B — They have delayed gastric emptying

Diabetic patients have delayed gastric emptying, therefore premedication against aspiration should be given.

Topic: Diabetes Mellitus – Premedication | Ref: Ch 23, Page 206

Q3. Answer: C — 24–48 hours prior to surgery

Due to long half-lives, sulfonyl urea and metformin should be stopped 24–48 hours prior to surgery.

Topic: Diabetes Mellitus – Management of Sugar | Ref: Ch 23, Page 206

Q4. Answer: C — 1/2 of the AM dose

The most widespread traditionally employed regime is to give 1/2 of the AM dose of insulin on the morning of surgery after starting 5% dextrose infusion.

Topic: Diabetes Mellitus – Insulin Management | Ref: Ch 23, Page 206

Q5. Answer: C — 125 mL/hour

5% dextrose is infused at a rate of 125 mL/hour to avoid the possibility of hypoglycemia.

Topic: Diabetes Mellitus – Insulin Management | Ref: Ch 23, Page 206

Q6. Answer: C — 25–30 mg/dL

1 unit of insulin decreases blood sugar levels by 25–30 mg/dL.

Topic: Diabetes Mellitus – Insulin Management | Ref: Ch 23, Page 206

Q7. Answer: B — Plasma glucose (mg/dL) / 150

Insulin is given by the formula: Units/hour = plasma glucose (mg/dL)/150.

Topic: Diabetes Mellitus – Insulin Infusion | Ref: Ch 23, Page 206

Q8. Answer: C — 150–180 mg/dL

The target should be to keep plasma sugar level between 150 and 180 mg/dL.

Topic: Diabetes Mellitus – Glucose Monitoring | Ref: Ch 23, Page 206

Q9. Answer: C — Myocardial ischemia

Myocardial ischemia is the most common cause of death in the perioperative period in diabetic patients.

Topic: Diabetes Mellitus – Monitoring | Ref: Ch 23, Page 206

Q10. Answer: B — Due to autonomic neuropathy

The MI may go silent in postoperative period if the patient has autonomic neuropathy.

Topic: Diabetes Mellitus – Monitoring | Ref: Ch 23, Page 206

Q11. Answer: B — Patients not having peripheral and/or autonomic neuropathy

Patients not having peripheral and/or autonomic neuropathy can safely receive regional anesthesia.

Topic: Diabetes Mellitus – Choice of Anesthesia | Ref: Ch 23, Page 207

Q12. Answer: B — Due to involvement of atlanto-occipital and temporomandibular joint (stiff joint syndrome)

Difficult intubation may be anticipated due to involvement of atlanto-occipital and temporomandibular joint (stiff joint syndrome).

Topic: Diabetes Mellitus – General Anesthesia | Ref: Ch 23, Page 207

Q13. Answer: C — Intravenous adrenaline

Hypotension and bradycardia in patients with autonomic neuropathy may not respond to atropine and ephedrine; intravenous adrenaline may act as savior.

Topic: Diabetes Mellitus – General Anesthesia | Ref: Ch 23, Page 207

Q14. Answer: B — Ketamine, Pancuronium, Atropine, Desflurane >6%

Drugs causing sympathetic stimulation like ketamine, pancuronium, atropine, desflurane >6% should be avoided as they cause hyperglycemia.

Topic: Diabetes Mellitus – General Anesthesia | Ref: Ch 23, Page 207

Q15. Answer: B — After the patient is euthyroid

Elective surgery should be deferred till the patient is euthyroid.

Topic: Hyperthyroidism | Ref: Ch 23, Page 207

Q16. Answer: B — Sleeping pulse rate

Sleeping pulse rate should be brought down to normal with beta blockers before taking the patient for surgery.

Topic: Hyperthyroidism – Preoperative Evaluation | Ref: Ch 23, Page 207

Q17. Answer: B — Large thyroid can produce tracheal compression

As large thyroid can produce tracheal compression, airway assessment by indirect laryngoscopy, X-ray neck or CT scan is must.

Topic: Hyperthyroidism – Preoperative Evaluation | Ref: Ch 23, Page 207

Q18. Answer: B — They can cause sympathetic stimulation

Regional anesthesia can be safely given in hyperthyroid patients however local anesthetics with adrenaline must not be used as they cause sympathetic stimulation.

Topic: Hyperthyroidism – Choice of Anesthesia | Ref: Ch 23, Page 207

Q19. Answer: B — They block the sympathetic response

Central neuraxial blocks are preferred because of their capacity to block the sympathetic response.

Topic: Hyperthyroidism – Choice of Anesthesia | Ref: Ch 23, Page 207

Q20. Answer: B — Thiopentone – possesses antithyroid property

As thiopentone possesses antithyroid property, it is the induction agent of choice for hyperthyroid patients.

Topic: Hyperthyroidism – General Anesthesia | Ref: Ch 23, Page 207

Q21. Answer: C — Isoflurane

As hyperthyroid patients are more prone for arrhythmias, most cardiac stable i.e. Isoflurane should be the preferred agent.

Topic: Hyperthyroidism – General Anesthesia | Ref: Ch 23, Page 207

Q22. Answer: C — No, MAC remains the same

Minimum alveolar concentration (MAC) of inhalational agents remains same in hyperthyroidism.

Topic: Hyperthyroidism – General Anesthesia | Ref: Ch 23, Page 207

Q23. Answer: B — A life-threatening condition most often occurring in postoperative period

Thyroid storm is a life-threatening condition most often occurring in postoperative period, however can occur intraoperatively also.

Topic: Hyperthyroidism – Postoperative Complications | Ref: Ch 23, Page 207

Q24. Answer: B — Dexamethasone

Management of thyroid storm includes dexamethasone which blocks conversion of T4 to T3.

Topic: Hyperthyroidism – Thyroid Storm | Ref: Ch 23, Page 207

Q25. Answer: B — Tracheal compression by expanding hematoma

Upper airway obstruction is most often due to tracheal compression caused by expanding hematoma, treated immediately by opening the sutures.

Topic: Hyperthyroidism – Postoperative Complications | Ref: Ch 23, Page 207-208

Q26. Answer: B — Hypocalcemia leading to laryngospasm

Hypocalcemia caused by hypoparathyroidism can precipitate laryngospasm.

Topic: Hyperthyroidism – Postoperative Complications | Ref: Ch 23, Page 208

Q27. Answer: B — IV esmolol or propranolol to settle tachycardia

At least tachycardia should be settled with IV esmolol or IV propranolol. Propranolol has the added advantage of inhibiting conversion of T4 to T3.

Topic: Emergency Surgery in Hyperthyroid Patient | Ref: Ch 23, Page 208

Q28. Answer: B — No, mild hypothyroidism should not be a deterrent to postpone the surgery

Ideally elective surgery should be deferred till euthyroid, however mild hypothyroidism should not be a deterrent to postpone the surgery.

Topic: Hypothyroidism | Ref: Ch 23, Page 208

Q29. Answer: B — Congestive cardiac failure, anemia, hypothermia, hyponatremia, hypoglycemia

Associated conditions like CCF, anemia, hypothermia, hyponatremia, and hypoglycemia should be ruled out.

Topic: Hypothyroidism – Preoperative Evaluation | Ref: Ch 23, Page 208

Q30. Answer: C — These patients are extremely sensitive to depressant drugs

Hypothyroid patients are extremely sensitive to depressant drugs therefore premedication with benzodiazepines should be avoided.

Topic: Hypothyroidism – Preoperative Evaluation | Ref: Ch 23, Page 208

Q31. Answer: B — Propofol
Induction is normally done with propofol; however, if there is evidence of low cardiac output then ketamine is preferred.

Topic: Hypothyroidism – General Anesthesia | Ref: Ch 23, Page 208

Q32. Answer: B — Mivacurium, Atracurium/Cis-atracurium
As mivacurium, atracurium/Cis-atracurium do not have the possibility of accumulation, they are preferred muscle relaxants in hypothyroidism.

Topic: Hypothyroidism – General Anesthesia | Ref: Ch 23, Page 208

Q33. Answer: C — The metabolism of drugs is reduced in hypothyroidism
As the metabolism of drugs is reduced in hypothyroidism, they should be used in minimal possible doses.

Topic: Hypothyroidism – General Anesthesia | Ref: Ch 23, Page 208

Q34. Answer: B — IV T3 and T4, and steroid cover
IV T3 and T4 should be given, and as adrenal function is often decreased, steroid cover also becomes a must.

Topic: Emergency Surgery in Hypothyroid Patient | Ref: Ch 23, Page 208

Q35. Answer: C — Phenoxybenzamine (alpha blocker)
Patient's hypertension must be stabilized by alpha blocker; most often used is phenoxybenzamine.

Topic: Pheochromocytoma | Ref: Ch 23, Page 208

Q36. Answer: B — Unopposed alpha action can cause malignant hypertension
Beta blockers must not be started before initiating therapy with alpha blocker otherwise unopposed action can cause malignant hypertension.

Topic: Pheochromocytoma | Ref: Ch 23, Page 208

Q37. Answer: B — It sensitizes myocardium to catecholamines
Halothane is absolutely contraindicated because it sensitizes myocardium to catecholamines.

Topic: Pheochromocytoma | Ref: Ch 23, Page 209

Q38. Answer: B — Profound hypertension and life-threatening arrhythmias due to increased catecholamines
Increase in catecholamines during tumor handling can cause profound hypertension and life-threatening arrhythmias.

Topic: Pheochromocytoma | Ref: Ch 23, Page 209

Q39. Answer: C — Hypotension
Hypotension is the most common cause of death in immediate postoperative period in pheochromocytoma patients, caused by abrupt withdrawal of catecholamines after ligation of tumor vein.

Topic: Pheochromocytoma | Ref: Ch 23, Page 209

Q40. Answer: C — Dextrose normal saline
Dextrose normal saline should be started immediately after vein ligation to prevent hypotension and hypoglycemia.

Topic: Pheochromocytoma | Ref: Ch 23, Page 209

Q41. Answer: B — 1–2 days
Hydrocortisone (100 mg/day) is continued for 1–2 days for unilateral and 3–6 days for bilateral adrenalectomy.

Topic: Pheochromocytoma | Ref: Ch 23, Page 209

Q42. Answer: B — Hypertension, hyperglycemia, volume overload, and hypokalemia
Hypertension, hyperglycemia, volume overload and hypokalemia must be corrected before elective surgery.

Topic: Cushing Syndrome | Ref: Ch 23, Page 209

Q43. Answer: C — Exogenous steroids
Exogenous steroids are the most common cause of secondary adrenal insufficiency.

Topic: Glucocorticoid Deficiency | Ref: Ch 23, Page 209

Q44. Answer: C — 3 weeks
Any patient who has taken steroid for more than 3 weeks in the last 1 year must receive intraoperative supplementation with hydrocortisone.

Topic: Glucocorticoid Deficiency | Ref: Ch 23, Page 209

Q45. Answer: B — 100 mg 8 hourly
100 mg of hydrocortisone 8 hourly beginning on the day of surgery and continued for 24–48 hours; continuous infusion at 10 mg/hour is considered better.

Topic: Glucocorticoid Deficiency | Ref: Ch 23, Page 209

Q46. Answer: B — It causes adrenocortical suppression
Etomidate is contraindicated as it causes adrenocortical suppression.

Topic: Glucocorticoid Deficiency | Ref: Ch 23, Page 209

Q47. Answer: B — It produces fluoride that can worsen nephropathy
Sevoflurane should be avoided if there is hypokalemic nephropathy due to its fluoride production.

Topic: Mineralocorticoid Excess | Ref: Ch 23, Page 209

Q48. Answer: B — Difficult airway management
The most important concern for acromegaly patients is difficult airway management due to prognathism, macroglossia, overgrowth of epiglottis, and narrowed glottis.

Topic: Pituitary Dysfunction – Acromegaly | Ref: Ch 23, Page 209

Q49. Answer: C — 1 year
Recovery of HPA takes 1 year after exogenous steroid use.

Topic: Glucocorticoid Deficiency | Ref: Ch 23, Page 209

Q50. Answer: B — Hypoxia
Hypothyroid patients can go into hypoxia in postoperative period due to hypoventilation.

Topic: Hypothyroidism – Postoperative | Ref: Ch 23, Page 208